Lab Worksheet 1: Computation

CORE 103 Computational Thinking

Fall 2026

Instructions

- This LW must be submitted online via CANVAS. You are required to submit a zip file that contains all the *.py* files.
- The zip file should be named as *LW1_aa123456.zip* where *aa123456* will be replaced with your student id.
- **Files that don't follow the appropriate naming convention will not be graded.**

Exercises

1. True/False Opposite Score

This question should be implemented in the file ***q1.py****.*

Ayan and Bilal are practicing for a short online true-or-false quiz. The quiz has N questions, each worth 1 mark, and there is no negative marking.

To make the practice interesting, Bilal decides that he will mark the opposite answer to Ayan on every question. If Ayan marks True, Bilal marks False; if Ayan marks False, Bilal marks True. After the quiz is checked, Ayan's score is K.

Given N and K, determine Bilal's score.

Constraints

- $N, K \in \mathbb{Z}$
- $1 \leq N \leq 100$
- $0 \leq K \leq N$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of N and K respectively.

The output must contain a single number denoting Bilal's score.

Sample

Input	Output
1 1	0
5 2	3

In the first case, $(N, K) = (1, 1)$. There is only 1 question in the quiz, and Ayan scores 1 mark, so Ayan answered it correctly. Since Bilal's answer is opposite, Bilal answered it incorrectly and scores 0 marks.

In the second case, $(N, K) = (5, 2)$. There are 5 questions in the quiz, and Ayan scores 2 marks, so Ayan answered 2 questions correctly and 3 incorrectly. Since Bilal's answers are opposite, Bilal answers those 2 questions incorrectly and the other 3 correctly. Therefore, Bilal scores 3 marks.

Exercise

Indicate the outputs for the given inputs in **q1.py**.

Input	Output
50 0	
100 50	
25 12	

Pseudocode

Implement the pseudocode in **q1.py**.

2. Quiz Score Tracker

*This question should be implemented in the file **q2.py**.*

A student takes a 10-question online quiz. The quiz is worth a total of T points, and every question carries the same number of points.

After submission, the system reports that the student answered C questions correctly. The student wants to know the final score shown by the quiz system.

Given T and C, determine the student's score.

Constraints

- $T, C \in \mathbb{Z}$
- $10 \leq T \leq 200$
- $0 \leq C \leq 10$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of T and C respectively.

The output must contain a single float denoting the corresponding score earned by the student.

Sample

Input	Output
20 3	6.0
15 5	7.5

In the first case, $(T, C) = (20, 3)$. There are 20 points for 10 questions. Each question is therefore worth 2 points. As the student got 3 questions correct, the score is $3 \times 2 = 6.0$.

In the second case, $(T, C) = (15, 5)$. There are 15 points for 10 questions. Each question is therefore worth 1.5 points. As the student got 5 questions correct, the score is $5 \times 1.5 = 7.5$.

Exercise

Indicate the outputs for the given inputs in **q2.py**.

Input	Output
10 3	
100 10	
130 4	

Pseudocode
Implement the pseudocode in **q2.py**.

3. Smoothie Ingredient Packs

*This question should be implemented in the file **q3.py**.*

A student-run smoothie stall is planning one weekly batch for a campus fundraiser. The batch uses two ingredients: smoothie base and fruit mix.

One base pack provides b milliliters of smoothie base, and one fruit pack provides f grams of fruit mix. For the weekly batch, the stall needs exactly B milliliters of smoothie base and F grams of fruit mix. The pack sizes are chosen so the required amounts can be measured using whole packs.

Determine the total number of ingredient packs required for the week.

Constraints

- $b, f, B, F \in \mathbb{Z}$
- $100 \leq b \leq 200$
- $400 \leq f \leq 500$
- $1000 \leq B \leq 1200$
- $1000 \leq F \leq 1500$
- B is divisible by b
- F is divisible by f

Interaction

The input comprises a single line containing 4 space-separated integers denoting the values of b, f, B, and F, respectively.

The output must contain a single number denoting the total number of ingredient packs required for the week.

Sample

Input	Output
100 400 1000 1200	13
150 400 1200 1200	11

In the first case, $b = 100$, $f = 400$, $B = 1000$, and $F = 1200$. Base packs contain 100 milliliters of smoothie base each, and fruit packs contain 400 grams of fruit mix each. The weekly batch needs 1000 milliliters of smoothie base and 1200 grams of fruit mix. Therefore, you require $\frac{1000}{100} = 10$ base packs and $\frac{1200}{400} = 3$ fruit packs. So a total of 13 ingredient packs are required per week.

In the second case, $b = 150$, $f = 400$, $B = 1200$, and $F = 1200$. Base packs contain 150 milliliters of smoothie base each, and fruit packs contain 400 grams of fruit mix each. The

weekly batch needs 1200 milliliters of smoothie base and 1200 grams of fruit mix. Therefore, you require $\frac{1200}{150} = 8$ base packs and $\frac{1200}{400} = 3$ fruit packs. So a total of 11 ingredient packs are required per week.

Exercise

Indicate the outputs for the given inputs in **q3.py**.

Input	Output
100 450 1000 1350	
200 500 1200 1500	
150 500 1050 1500	

Pseudocode

Implement the pseudocode in **q3.py**.

4. Poster Frame Cost

*This question should be implemented in the file **q4.py**.*

A student is preparing a rectangular project poster for display day. To make it look neat, the student wants to place a wire frame around the outer boundary of the poster.

The poster has length L cm and width W cm. The wire is sold by length, and each centimeter of wire costs Rs. C.

Given L, W, and C, determine the total cost of the frame.

Constraints

- $L, W, C \in \mathbb{Z}$
- $1 \leq L, W, C \leq 1000$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of L, W, and C respectively.

The output must contain a single number denoting the total cost of the wire frame.

Sample

Input	Output
10 10 10	400
1 2 3	18

In the first case, $(L, W, C) = (10, 10, 10)$. The length of the wire frame is 10cm and the width is 10cm. The total perimeter is $2 \cdot 10 + 2 \cdot 10 = 40$cm. Since the cost is 10 rupees per cm, the total cost is $40 \cdot 10 = 400$ rupees.

In the second case, $(L, W, C) = (1, 2, 3)$. The length of the wire frame is 1cm and the width is 2cm. The total perimeter is $2 \cdot 1 + 2 \cdot 2 = 6$cm. Since the cost is 3 rupees per cm, the total cost is $6 \cdot 3 = 18$ rupees.

Exercise

Indicate the outputs for the given inputs in **q4.py**.

Input	Output
11 15 50	
1000 1000 1000	
67 37 23	

Pseudocode
Implement the pseudocode in **q4.py**.

5. Assignment Deadline Countdown

*This question should be implemented in the file **q5.py**.*

A course project is assigned with a deadline after W weeks. Several days have already passed since the project was announced.

The class representative is preparing a reminder message and knows that D days have passed so far. The reminder should mention how many days are still left before the deadline.

Given W and D, determine how many days remain until the deadline.

Constraints

- $W, D \in \mathbb{N}$
- $1 \leq W \leq 10$
- $1 \leq D < 7W$

Interaction

The input comprises a single line containing 2 space-separated integers denoting the values of W and D.

The output must contain a single number denoting the number of days remaining until the deadline.

Sample

Input	Output
1 5	2
2 13	1

In the first case, $(W, D) = (1, 5)$. The project is due after 1 week, and 5 days have already passed. 2 days remain until the deadline.

In the second case, $(W, D) = (2, 13)$. The project is due after 2 weeks, and 13 days have already passed. 1 day remains until the deadline.

Exercise

Indicate the outputs for the given inputs in **q5.py**.

Input	Output
3 16	
5 7	
6 12	

Pseudocode
Implement the pseudocode in **q5.py**.

6. Arcade Token Exchange

*This question should be implemented in the file **q6.py**.*

At a campus arcade, game tokens are bought using coins. A student has F coins worth Rs. 5 each and T coins worth Rs. 10 each.

One game token costs Rs. C. The arcade machine only gives whole tokens, so any leftover money that is not enough for another token cannot be used.

Given F, T, and C, determine the maximum number of tokens the student can buy.

Constraints

- $F, T, C \in \mathbb{Z}$
- $1 \leq F, T, C \leq 1000$

Interaction

The input comprises a single line containing 3 space-separated integers denoting the values of F, T, and C respectively.

The output must contain a single number denoting the number of game tokens the student can buy.

Sample

Input	Output
10 10 10	15
3 1 8	3

In the first case, $(F, T, C) = (10, 10, 10)$. The student has 10 coins worth Rs. 5 each and 10 coins worth Rs. 10 each. This gives Rs. 150 in total. Since each game token costs Rs. 10, the student can buy 15 tokens.

In the second case, $(F, T, C) = (3, 1, 8)$. The student has 3 coins worth Rs. 5 each and 1 coin worth Rs. 10. This gives Rs. 25 in total. Since each game token costs Rs. 8, the student can buy 3 tokens.

Exercise

Indicate the outputs for the given inputs in **q6.py**.

Input	Output
4 4 1000	
10 5 1	
13 31 35	

Pseudocode

Implement the pseudocode in **q6.py**.

7. Robot Grid Journey

*This question should be implemented in the file **q7.py**.*

A classroom robot is being tested on a tiled grid. It starts at position `(rs, cs)` and must reach position `(rd, cd)`.

In one command, the robot always moves to the next row. From its current position `(r, c)`, it can either move diagonally to `(r + 1, c + 1)` or move straight down to `(r + 1, c)`. The destination is chosen so that at least one valid path exists.

Determine the number of commands required to reach the destination.

Constraints

- `rs`, `cs`, `rd`, and `cd` are integers.
- Each value is between `0` and `1000`, inclusive.

- There exists one or more valid paths from `(rs, cs)` to `(rd, cd)` using the allowed moves.

Interaction

The input comprises a single line containing 4 space-separated integers denoting the values of `rs`, `cs`, `rd`, and `cd`, respectively.

The output must contain a single number denoting the number of commands the robot must make.

Sample

Input	Output
0 0 5 5	5
4 4 11 10	7

In the first case, the robot starts at `(0, 0)` and must reach `(5, 5)`. It can use 5 diagonal moves of the type `(r + 1, c + 1)`.

In the second case, the robot starts at `(4, 4)` and must reach `(11, 10)`. It can use 6 diagonal moves of the type `(r + 1, c + 1)` and 1 straight move of the type `(r + 1, c)`.

Exercise

Indicate the outputs for the given inputs in **q7.py**.

Input	Output
0 0 7 4	
3 2 10 9	
9 3 17 11	

Pseudocode

Implement the pseudocode in **q7.py**.